Lab 4 - Revolve - Funnel from a 2D Profile

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 01: Overview of Additive Manufacturing and 3D Scanning TSC objective: A1 - apply the revolve technique and choose a print orientation that avoids supports

Goal

Model a funnel by revolving a half-profile, then invert, slice and print it.

What you'll build

A working funnel revolved from a single 2D profile, printed wide-end down.

Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

Steps

1. Sketch a closed 2 mm funnel half-profile on the XZ plane: wide cone to narrow spout.

```
Create Sketch > Line - cone Ø 60 mm -> spout Ø 8 mm, wall 2 mm
```

2. Revolve the profile 360° around the vertical (Z) axis to form the funnel body.

```
Create > Revolve - profile: half section, axis: Z, angle 360 degrees
```

3. Use Inspect > Section Analysis to confirm a watertight, uniform wall.

```
Inspect > Section Analysis
```

4. Export the funnel as an STL file.

```
File > Export > STL
```

5. Open it in Cura - it imports spout-up; rotate it so the wide mouth sits on the build plate.

```
Cura > Rotate - 180 degrees  about X (wide end down)
```

6. Slice and confirm the cone angle prints without supports, then print.

```
Cura > Slice > Preview > Save to Disk
```

Evidence to capture

- Fusion 360 screenshot of the closed half-profile and 360-degree revolve.
- Cura Preview screenshot showing wide-end-down, support-free orientation.
- Photo or water-flow note confirming a clear spout.

Test it

Sliced upside-down the funnel needs no supports; water poured into the printed funnel flows cleanly through the spout.